

# CS 651 Group 7

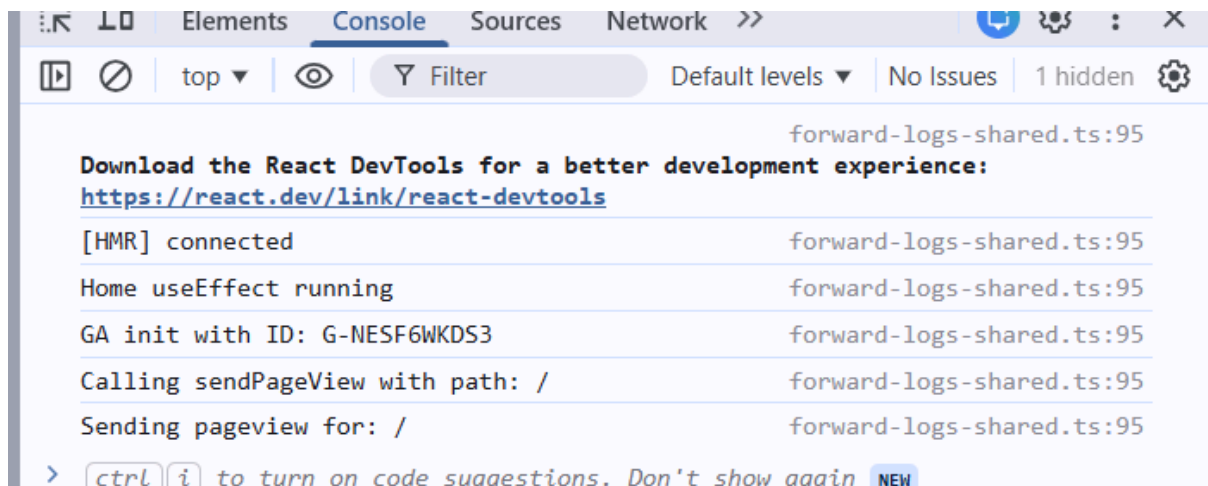
## Project 2 : Analytics and logging Report

Members

Jolene Le , Rohan Kumar, Vanessa Rodriguez, Shiva Sharan

### SECTION 1 : Google Analytics

We implemented analytics on both the client and server sides of the NewMe app. On the frontend, I integrated Google Analytics 4 using the `react-ga4` library: I initialize GA when the app loads, send a `page_view` event on route changes, and fire custom events for key interactions like the “Try it Now” button, each Gemini hairstyle generation in the editor, and when a user saves a favorite look to their gallery. On the backend, I added Google Cloud Logging to the Express server so every transformation-related API call (save, list, stats, delete) is logged with fields such as route, userId, status, and duration, which lets us analyze feature usage and performance over time. For this report, the three main metrics we focus on are: (1) the number of Try It Now CTA clicks (how many users start the style journey), (2) the number of Gemini hair-style generation requests (how often the core AI feature is used), and (3) the number of “Save favorite look” events (how many transformations users like enough to keep).



```
import ReactGA from 'react-ga4';

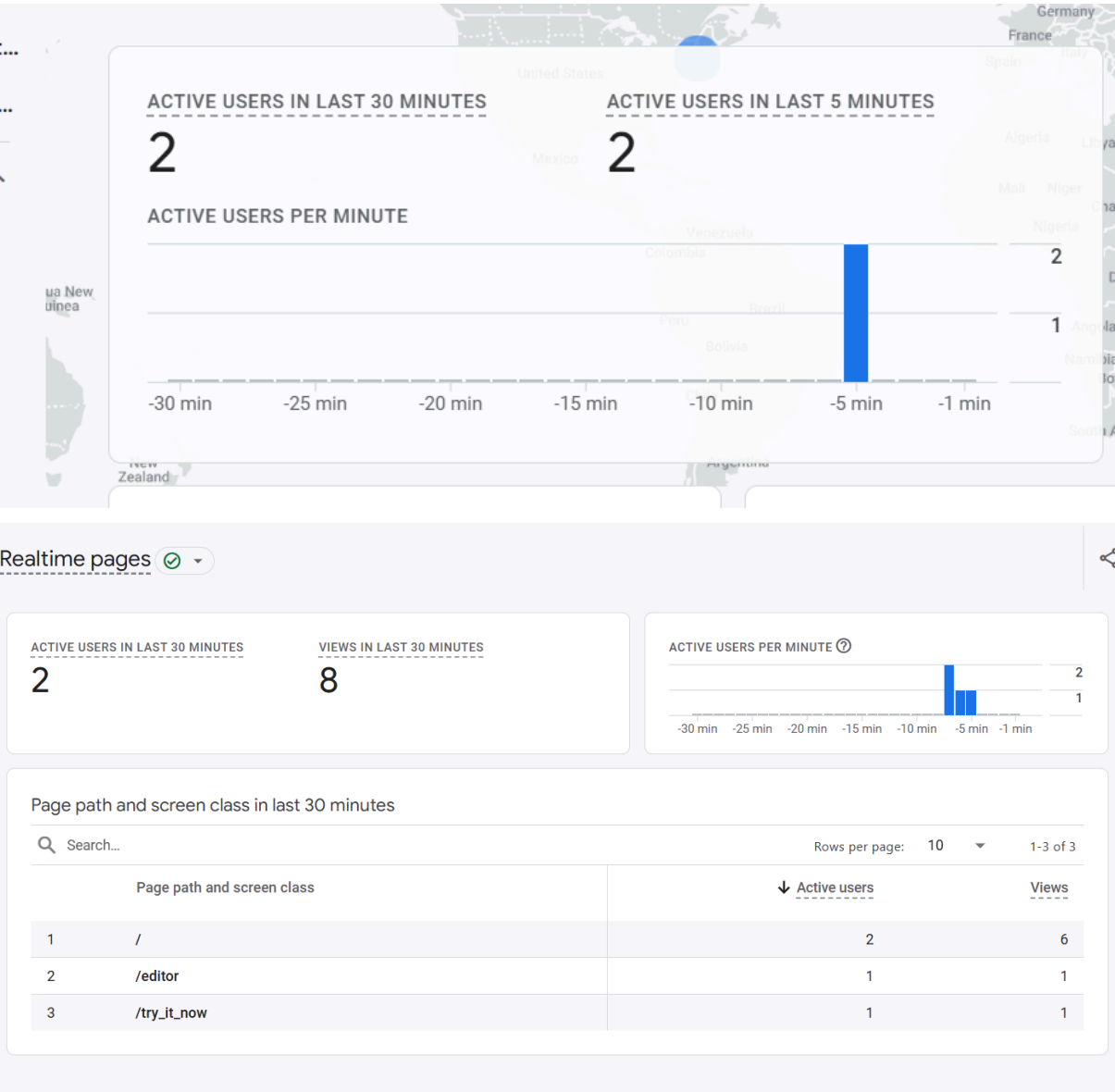
// Replace this later with your real GA4 ID, like "G-ABCD1234"
const GA_MEASUREMENT_ID = "G-NESF6WKDS3";

export function initGA() {
  ReactGA.initialize(GA_MEASUREMENT_ID);
}

export function sendPageView(path) {
  ReactGA.send({
    hitType: "pageview",
    page: path,
  });
}
```

1.1 Metric 1:Try It Now button clicks

1.1a Visualization



1.1b Interpretation

This metric shows how many times users click the main “Try It Now” button on the homepage. When the number is higher, it means more visitors are interested in actually trying the app, not just looking at the page. Spikes in the graph usually line up with times when we shared the link or showed the app to friends or classmates. If the numbers stay low, it suggests the homepage is not convincing enough or we are not getting much traffic. Overall, this metric tells us how effective our homepage is at turning visitors into active users who start a style transformation.

### 1.1c Limitations

This metric counts clicks, not unique people, so one user clicking several times can make the number look bigger.

It only tracks users who allow Google Analytics and JavaScript, so some clicks may be missing from the data.

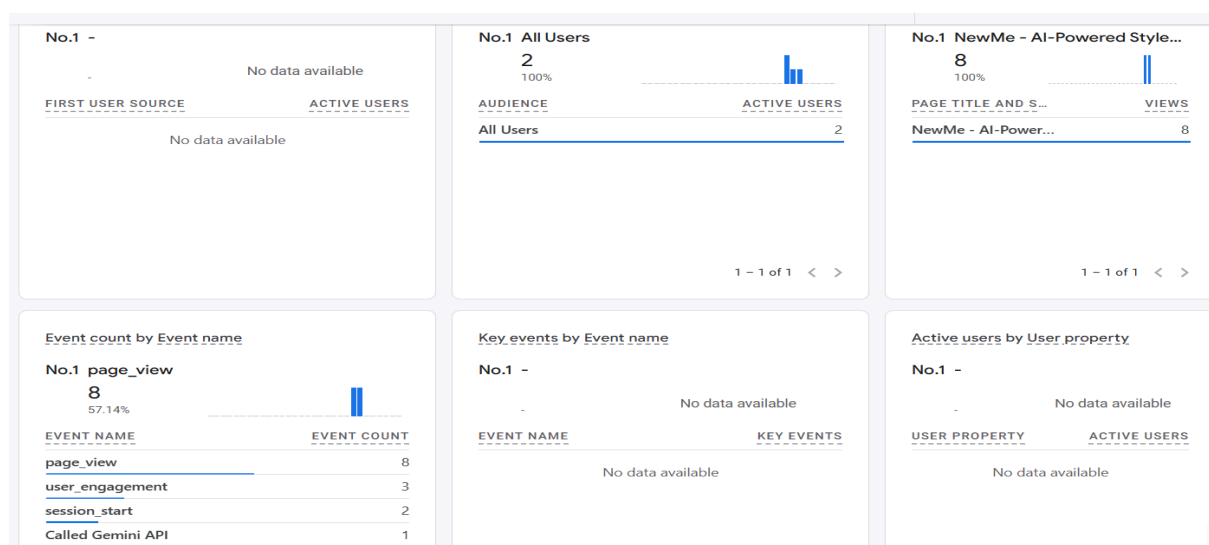
It doesn’t tell us what happens after the click (for example, if the user actually completes a transformation or quits).

Because this is a small class project, the total number of clicks is low, so trends can be hard to interpret.

On its own, this metric doesn’t show full user behavior, so we need to combine it with other metrics like saves or backend logs.

## 1.2 Metric 2: AI transformation requests

### 1.2.a Visualization



### 1.2.b Interpretation

This metric counts how many times users ask the AI to generate a new hairstyle or hair color.

If the number is high, it means users are actively experimenting and trusting the AI to change their look.

Spikes in this metric after certain days or demos can show when users are most curious or engaged.

If many people click “Try It Now” but few trigger the AI, it suggests confusion or friction in the flow.

Overall, this metric tells us how much people are using the “smart” core feature of our app, not just browsing.

### **1.2.c Limitations**

This metric only shows **how many** AI requests were made, not whether the results were good or bad.

One user can trigger the AI many times in a row, so the numbers don’t always equal the number of unique, happy users.

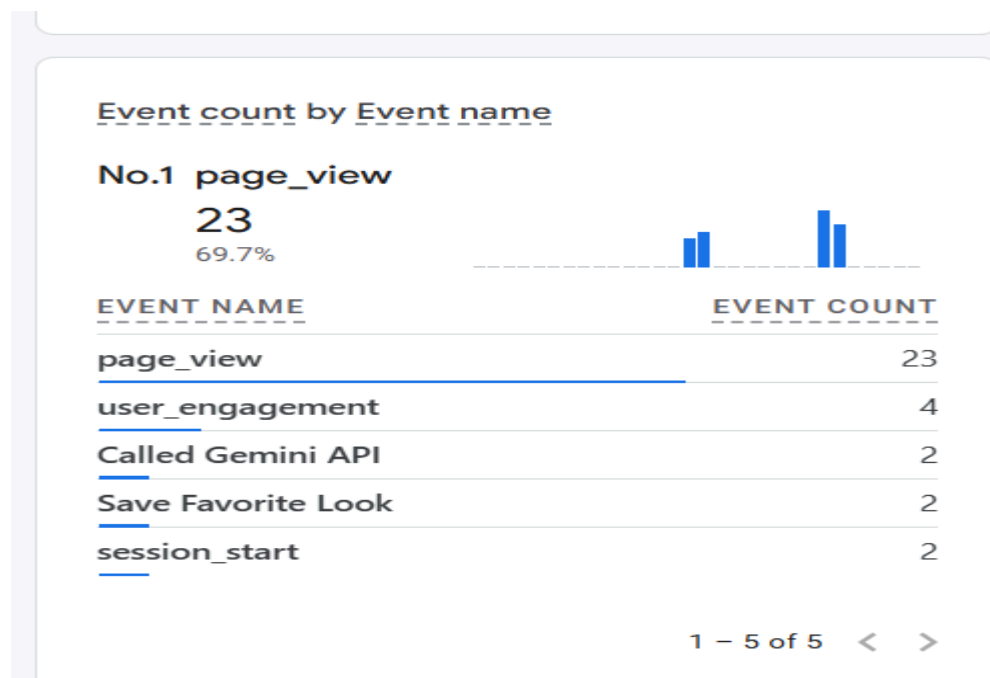
If the AI is slow or fails, we still might count the event, so the metric can overestimate real successful transformations.

We also don’t see the context — for example, whether people stop using the app right after a bad AI result.

Because our user base is small, a few heavy testers or group members can strongly change the trend and make it look bigger than it really is.

## **1.3 Metric 3: Save to Gallery / Save Favorite Look**

### **1.3.a Visualization**



UI8j8Q3StTRbwRc3X1Wf

+ Start collection

+ Add field

createdAt: 30 November 2025 at 14:46:03 UTC-8

hairColor: "Silver"

hairStyle: "Curly Layers"

originalImageUrl: "/api/photos/proxy-image?url=https%3A%2F%2Flh3.googleusercontent.com%2FgdTb45RTyPLX792b2Eto59L4Oqte3bbx42GgmUE9rh512"

prompt: "Modify ONLY the hair in this image. Change the hair color to: Silver Change the hairstyle to: Curly Layers PRESERVE EXACTLY: - All facial features (eyes, nose, mouth, face shape) - Skin tone, makeup, and facial hair (unless specified to change) - Expression and emotion - Clothing, accessories, and jewelry - Background and environment - Lighting, shadows, and photo quality - Camera angle and composition - Body position and pose REQUIREMENTS: - Generate a photorealistic result - Make the new hairstyle look natural and realistic - Match the original image quality and resolution"

### 1.3.b Interpretation

This metric shows how many times users save a transformed photo to their gallery as a favorite look.

Higher numbers mean users are not just testing the tool, but actually like the results enough to keep them.

If we see many "Try It Now" clicks but few saves, it suggests users are curious but not satisfied with the transformations.

If saves increase over time, it can mean our hairstyle and color options are useful and feel realistic to users.

Overall, this metric tells us how engaged users are with the final results and whether the app is delivering real value.

### 1.3.c Limitations

This metric counts save actions, but it doesn't tell us why users liked or didn't like a specific transformation. One user might save many looks, while another saves none, so the numbers don't directly equal the number of happy users. If a user screenshots the page instead of using "Save to Gallery," that behavior will not be tracked in this metric. Because the project has a small user base, a few very active users can heavily influence the graph. This metric also doesn't show any long-term behavior, like whether users come back later to view or reuse their saved looks.

## Section 2: Google Cloud Logging

### Summary of Implementation

I set up Google Cloud Logging on the backend using the `@google-cloud/logging` library. The Express server uses a shared `writeLog()` helper (in `server/index.ts`) that writes structured log entries to a log stream called `newme-app-logs` in the same Google Cloud project as our Firebase app.

Each time a user works with transformations (save to gallery, list gallery, view details, delete, or fetch stats), the server logs an entry with fields like `route`, `method`, `userId`, `durationMs`, and whether the request **succeeded or failed**. On the frontend, user actions such as **"Apply Transformation"** and **"Save to Gallery"** go through these backend routes, so each important UI action is recorded in Cloud Logging with useful metadata.

### 2.1 Frontend Implementation (User Actions → Logs)

Google Cloud NewMe Search (/) for resources, docs, products, and more

Logs Explorer

Query library Share link Preferences < unavailable > Learn

Project logs Search all fields

All resources All log names All severities Correlate by

Run query Show query

1 log\_name="projects/newme-477700/logs/cloudbuild" resource.type="build" resource.labels.build\_id="1d328a67-2c47-4468-bb17-d1d4a81ac0d6" timestamp>="2025-11-29T04:11:51.800051Z" timestamp<="2025-11-29T04:16:32.334Z"

Example queries Query language guide Language: LQL

Fields Search fields and values

System Metadata

Severity 188

Showing top 1 of 1 value

Info

BUILD ID 188

BUILD TRIGGER ID 188

Log name 188

Project ID 188

Resource type 188

JSON payload (most frequent) Preview

No results found

Timeline

20 11:54:000 20 12:30 20 13:00 20 13:30 20 14:00 20 14:30 20 15:00 20 15:30 20 16:00 20 16:30:000

188 results

| SEVERITY | TIME                    | SUMMARY                                                                                                                |
|----------|-------------------------|------------------------------------------------------------------------------------------------------------------------|
| >        | 2025-11-28 20:15:29.882 | Step #2 - "Deploy": Service URL: https://newme-server-236230481433.us-west1.run.app                                    |
| >        | 2025-11-28 20:15:29.609 | Finished Step #2 - "Deploy"                                                                                            |
| >        | 2025-11-28 20:15:29.652 | PUSH                                                                                                                   |
| >        | 2025-11-28 20:15:29.652 | Pushing us-west1-docker.pkg.dev/newme-477700/cloud-run-source-deploy/newme/newme-server:333eccd9b9f547e9f68d2f07d441f- |
| >        | 2025-11-28 20:15:30.163 | The push refers to repository [us-west1-docker.pkg.dev/newme-477700/cloud-run-source-deploy/newme/newme-server]        |
| >        | 2025-11-28 20:15:30.512 | a289849fdc96: Preparing                                                                                                |
| >        | 2025-11-28 20:15:30.512 | d7c667379ebf: Preparing                                                                                                |
| >        | 2025-11-28 20:15:30.512 | ae6304901fe: Preparing                                                                                                 |
| >        | 2025-11-28 20:15:30.512 | b597fe0bb5b: Preparing                                                                                                 |
| >        | 2025-11-28 20:15:30.512 | b0643b6f6505: Preparing                                                                                                |
| >        | 2025-11-28 20:15:30.512 | fbaeb92e8a42: Preparing                                                                                                |
| >        | 2025-11-28 20:15:30.512 | 6acabb84a1c3: Preparing                                                                                                |
| >        | 2025-11-28 20:15:30.512 | 256f393e029f: Preparing                                                                                                |
| >        | 2025-11-28 20:15:30.697 | 256f393e029f: Layer already exists                                                                                     |
| >        | 2025-11-28 20:15:30.967 | b0643b6f6505: Layer already exists                                                                                     |
| >        | 2025-11-28 20:15:30.969 | b597fe0bb5b: Layer already exists                                                                                      |
| >        | 2025-11-28 20:15:30.970 | 6acabb84a1c3: Layer already exists                                                                                     |
| >        | 2025-11-28 20:15:30.972 | d7c667379ebf: Layer already exists                                                                                     |
| >        | 2025-11-28 20:15:30.973 | a289849fdc96: Layer already exists                                                                                     |
| >        | 2025-11-28 20:15:30.983 | fbaeb92e8a42: Layer already exists                                                                                     |
| >        | 2025-11-28 20:15:30.983 | ae6304901fe: Layer already exists                                                                                      |
| >        | 2025-11-28 20:15:31.508 | 333eccd9b9f547e9f68d2f07d441f592520ced9d: digest: sha256:184176e15702d2e511d95951ce6b7095380b41d3747ef4e9414470b49f1c- |
| >        | 2025-11-28 20:15:31.725 | DONE                                                                                                                   |

Showing logs for time specified in query. To view more results update your query.

## 2.2 Backend Implementation (Express / NodeJS)

https://console.cloud.google.com/run/detail/us-west1/newme-server/observability/logs?project=newme-477700&supportedpurview=pro...

Google Cloud NewMe Search (/) for resources, docs, products, and more Search

Service details Edit & deploy new revision Edit repo settings Test Learn Refresh

newme-server Region: us-west1 URL: https://newme-server-236230481433.us-west1.run.app Scaling: Auto (Min: 0) Build History

Observability Revisions Source Triggers Networking Security YAML

Metrics Logs Severity: Default Filter Search all fields and values

| Severity | Timestamp                   | Summary                                                                                                                             |
|----------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| >        | 2025-11-29 14:15:25.467 PST | Firestore initialized                                                                                                               |
| >        | 2025-11-29 14:15:25.476 PST | Express server running on 0.0.0.0:8080                                                                                              |
| >        | 2025-11-29 14:15:25.476 PST | Health check: http://0.0.0.0:8080/health                                                                                            |
| >        | 2025-11-29 14:15:25.477 PST | Default STARTUP TCP probe succeeded after 1 attempt for container "placeholder-1" on port 8080.                                     |
| >        | 2025-11-29 14:15:32.142 PST | DELETE 200 336 B 389 ms node https://newme-server-236230481433.us-west1.run.app/api/transformations/4RYBQUuL0aIP18yLGmk2            |
| >        | 2025-11-29 14:15:32.532 PST | Transformation deleted: 4RYBQUuL0aIP18yLGmk2 by user 105476219578752713807                                                          |
| >        | 2025-11-29 14:15:33.573 PST | GET 200 879 B 175 ms node https://newme-server-236230481433.us-west1.run.app/api/transformations?limit=100                          |
| >        | 2025-11-29 14:16:13.645 PST | GET 200 879 B 163 ms node https://newme-server-236230481433.us-west1.run.app/api/transformations?limit=100                          |
| >        | 2025-11-29 14:16:21.320 PST | GET 200 879 B 127 ms node https://newme-server-236230481433.us-west1.run.app/api/transformations?limit=100                          |
| >        | 2025-11-29 14:16:44.687 PST | GET 200 879 B 126 ms node https://newme-server-236230481433.us-west1.run.app/api/transformations?limit=100                          |
| >        | 2025-11-30 05:55:08.384 PST | GET 404 467 B 2.1 s Chrome 131 https://newme-server-236230481433.us-west1.run.app/robots.txt                                        |
| >        | 2025-11-30 05:55:08.392 PST | Starting new instance. Reason: AUTOSCALING - Instance started due to configured scaling factors (e.g. CPU utilization, request t... |
| >        | 2025-11-30 05:55:09.610 PST | > social-network-v00.1.0 start:server                                                                                               |
| >        | 2025-11-30 05:55:09.610 PST | > node server/dist/index.js                                                                                                         |
| >        | 2025-11-30 05:55:10.682 PST | ENV DEBUG => {                                                                                                                      |
| >        | 2025-11-30 05:55:10.682 PST | FIREBASE_SERVICE_ACCOUNT: 'SET',                                                                                                    |
| >        | 2025-11-30 05:55:10.682 PST | FIREBASE_PROJECT_ID: 'newme-477700',                                                                                                |
| >        | 2025-11-30 05:55:10.682 PST | FIREBASE_STORAGE_BUCKET: 'newme-477700.firebaseiostorage.app',                                                                      |
| >        | 2025-11-30 05:55:10.682 PST | GOOGLE_CLIENT_ID: 'SET',                                                                                                            |
| >        | 2025-11-30 05:55:10.682 PST | GOOGLE_CLIENT_SECRET: 'SET'                                                                                                         |
| >        | 2025-11-30 05:55:10.682 PST | }                                                                                                                                   |
| >        | 2025-11-30 05:55:10.780 PST | Firebase Admin initialized successfully                                                                                             |
| >        | 2025-11-30 05:55:10.780 PST | Storage bucket: newme-477700.firebaseiostorage.app                                                                                  |
| >        | 2025-11-30 05:55:10.783 PST | Firestore initialized                                                                                                               |
| >        | 2025-11-30 05:55:10.792 PST | Default STARTUP TCP probe succeeded after 1 attempt for container "placeholder-1" on port 8080.                                     |
| >        | 2025-11-30 05:55:10.793 PST | Express server running on 0.0.0.0:8080                                                                                              |
| >        | 2025-11-30 05:55:10.793 PST | Health check: http://0.0.0.0:8080/health                                                                                            |
| >        | 2025-11-30 05:55:11.519 PST | GET 302 0 B 0 ms Chrome 131 http://newme-server-236230481433.us-west1.run.app/robots.txt                                            |
| >        | 2025-11-30 05:55:11.581 PST | GET 404 467 B 2 ms Chrome 131 https://newme-server-236230481433.us-west1.run.app/robots.txt                                         |

No newer entries found matching current filter.

```
// --- Google Cloud Logging setup ---
const logging = new Logging({
  projectId: serviceAccount.project_id, // from your service account JSON
  keyFilename: serviceAccountPath, // same file you already created
});

const appLog = logging.log('newme-app-logs'); // name of the log stream

type LogSeverity = 'DEBUG' | 'INFO' | 'WARNING' | 'ERROR';

export async function writeLog(
  severity: LogSeverity,
  message: string,
  data: Record<string, any> = {}
) {
  try {
    const metadata = {
      resource: { type: 'global' },
      severity,
    };

    const entry = appLog.entry(metadata, {
      message,
      ...data,
      timestamp: new Date().toISOString(),
    });

    await appLog.write(entry);
  } catch (err) {
    console.error('[LOGGING] Failed to write log entry:', err);
  }
}
```